

A1. CHAPTER 1 : INTRODUCTION(20 Marks) [CO1,PO1] & [CO1, PO2]

Question No		0 (Not Available)	1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
A1	Project Background (5 Marks)	Not Available	Project backgrounds are very weak, incomplete, and irrelevant.	Project backgrounds are incomplete, inaccurate, and confusing.	Project backgrounds are complete but not very relevant to the project work.	Project backgrounds are complete and have good relevancy to the project work.	Project backgrounds are highly relevant to the project work, complete, accurate and clearly stated.	1
A2	Problem Statements (5 Marks)	Not Available	Problem statements are severely underdeveloped, absence of focus; statements are very weak, incomplete, or irrelevant.	Problem statements are poorly developed, inaccurate, and statements are confusing.	Some shortcomings in clarity of the problem statements. Statements are presented but unclear, not very focused or inexplicit.	The problems in the project are identified and briefly explained. However, the connections to the objectives and scopes are somewhat unclear	The problems in the project are clearly identified, comprehensively explained, and directly linked to the objectives and scopes of the project.	1
A3	Objectives (5 Marks)	Not Available	Objectives are severely underdeveloped without focus. Statements are not complete or irrelevant.	Objectives are poorly developed and inaccurate. Statements are confusing	Objectives are developed, may not accurate and completely. The connections of the research gaps to the objectives are linked but	Objectives in the project are identified and presented. However, the connections to the problem and scopes are somewhat	Objectives are presented and focused. The connections of the research problems to the objectives are linked to the objectives and scopes.	1

					somewhat unclear or insignificant.	unclear.		
A4	Scopes / limitation (5 marks)	Not Available	Research scope is severely underdeveloped without focus. Statements very weak, incomplete, or irrelevant. Limitations of the work are not identified.	Research scope is developed however poorly and inaccurate, Statements is confusing. Limitations of the work are poorly identified.	Research scope is developed, not completely. Statements may not be complete and slightly inaccurate. Limitations of the work is mentioned briefly and not explained.	Research scope is clearly stated. are somewhat not linked to the objectives. Statements may be briefly. Limitations of the work is mentioned briefly and explained.	Research scope is clearly stated and are linked to the objectives. Statements are complete and accurate. Limitations of the work are identified and well explained.	1

B. CHAPTER 2: LITERATURE REVIEW (30 MARKS) [CO1, PO1], [CO1, PO2] & [CO2, PO4]

Question No.			1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
B1	Literature Review (5 Marks)	Not Available	Theory background is completely irrelevant to the project or unclear at all. Key of theory background challenges and motivation not identified	Theory background poorly surveyed. Key of theory background challenges and motivation not identified	Theory background briefly surveyed. Key of theory background challenges and motivation vaguely identified	Theory background broadly surveyed. Key theory background challenges identified and explained.	Theory background broadly surveyed and analysed. Key theory background challenges identified and well explained.	1
B2	Theoretical Review and Previous Work Review (5 Marks)	Not Available	The theoretical and previous work review is completely irrelevant to the project or unclear at all. Demonstrates a very	Most theoretical and previous work is missing, not clearly explained or irrelevant to the project. Demonstrates a limited understandin	The theoretical and previous work review is briefly explained, relevant to the project but some of the important	The theoretical and previous work review is comprehensively explained but some are irrelevant to	The theoretical and previous work review is comprehensively explained and very relevant to the project.	1

			limited understanding of the relevant aspects of the topics or the theme.	g of the relevant aspects of the topics or the theme.	theories are missing. Demonstrates a sufficient understanding of the relevant aspects of the topics or the theme.	the project. Demonstrates a good understanding of the relevant aspects of the topics or the theme.	Demonstrates a perceptive understanding of the relevant aspects of the topics or the theme.	
B3	Relevancy of the literature review (10 Marks)	Not Available	The supporting literature is completely irrelevant to the work and is reviewed very badly. The number of journal/conference papers found is insufficient to support the literature review.	The supporting literature is mostly irrelevant to the work and is reviewed badly. The number of journal/conference papers found is insufficient to support the literature review.	There are only a few supporting literatures which is only slightly relevant to the work and is reviewed inadequately. The number of journal/conference papers found is sufficient, but all sources are outdated (more than 5 years).	Most of the supporting literature is relevant and important to the research work and is reviewed well. The number of journal/conference papers found is sufficient, but not all sources are up-to date.	All the sources of the supporting literature are very relevant and important to the research work with critically reviewed. The number of journal/conference papers found is sufficient, all of the sources published within 5 years.	2

B4	Investigate and summarization of the related work towards solving engineering problem (10 Marks)	Not Available	The review displays very limited details and development of other work. The student's response to literature demonstrates little or no awareness of the author's intention or techniques and not able to compares and summarizes the impact of present work towards solving engineering problem. No gap of knowledge is included.	The review displays limited details and development of other work. The student's response to literature demonstrates little or no awareness of the author's intention or techniques and not able to compares and summarizes the impact of present work towards solving engineering problem. No gap of knowledge is included.	The review displays some illustrative details and development of other work. The student's response to literature sometimes demonstrates an awareness of the author's intention and techniques and able to compares and summarizes the impact of present work towards solving engineering problem. Gap of knowledge is not clearly defined.	The review consistently displays illustrative details and development of other work. The student's response to literature demonstrates an awareness of the author's intention and techniques and able to compares and summarizes the impact of present work towards solving engineering problem. Gap of knowledge is defined.	The review consistently displays illustrative details and development of other work in-depth analysis on the results and information obtained. The student's response to literature demonstrates an awareness of the author's intention and techniques and able to compares and summarizes the impact of present work towards solving engineering problem. Gap of knowledge is clearly defined.	2
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C. CHAPTER 3: METHODOLOGY (30 MARKS) [CO2, PO3]

Question No		0 (Not Available)	1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
C1	Appropriate tools and procedure approaches (15 Marks)	Not Available	Tools and procedures are vaguely identified.	Tools and procedures are clearly identified. Few appropriate references are used.	Tools and procedures are clearly identified and briefly explained.	Tools and procedures are clearly identified and comprehensively explained.	Tools and procedures are clearly identified, comprehensively explained and rationally justified.	3
C2	Complexity of the technique and design solution (15 Marks)	Not Available	Deal with only a single component or one design	Deal with very less components or design parameters	Deal with a few components or design parameters	Deal with some components or design parameters	Deal with many components or design parameters	3

			parameter with the solution shows no original thought or creativity to complete the project.	with the solution is lacking in creativity and originality to complete the project.	with approach may be similar to literature but with some modifications to complete the project.	with the solution is somewhat creative and original to complete the project (good values).	with the solution is creative and presents 100% originality to complete the project (excellent values).	
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D. REPORT STRUCTURE (15 MARKS) [CO3, PO10]

Question No		0 (Not Available)	1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
D1	Report organization (5 Marks)	No report available	Poorly structured with no clear progression through and between each sections.	Moderately structured with little progression through and between each sections.	Well-structured with acceptable progression through and between each sections.	Well-structured with good progression through and between each sections.	Excellent structured with clear and logical progression through and between each sections.	1

D2	Report writing (5 Marks)	No report available	Writing is incorrect and very hard to read, numerous spelling and grammatical errors.	Writing is somehow correct but not pleasant to read, numerous spelling and grammatical errors.	Writing is basically correct and readable, spelling and grammatical errors are present but at acceptable quantities.	Writing is correct and pleasant to read, some spelling and grammatical errors.	Writing is fluid and pleasant to read, No spelling errors and few grammatical errors.	1
D3	Figures, tables and references format (5 Marks)	No tables & figures included in report	Unacceptable used of figures, tables and related references	Poor used of figures, tables and related references	Adequate used of figures, tables and related references	Good used of figures, tables and related references (mostly within 5 years)	Excellent used of figures, tables and related references (mostly within 5 years)	1

A1. CHAPTER 3 : METHODOLOGY (40 Marks) [CO1,PO3] & [CO1, PO4]

Question No		0 (Not Available)	1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
A1	Appropriate tools and design procedure approaches (15 Marks)	Not Available	Tools and procedures are vaguely identified.	Tools and procedures are clearly identified. Few appropriate references are used.	Tools and procedures are clearly identified and briefly explained.	Tools and procedures are clearly identified and comprehensively explained.	Tools and procedures are clearly identified, comprehensively explained and rationally justified.	3
A2	Complexity of the design solution (10 Marks)	Not Available	Deal with only a single component or one design parameter with the solution shows no original thought or creativity to complete the project. The designed solution do not aligned with problem	Deal with very less components or design parameters with the solution is lacking in creativity and originality to complete the project. The designed solution fairly aligned with at least one problem	Deal with a few components or design parameters with approach may be similar to literature but with some modifications to complete the project. The solution aligned with more than	Deal with some components or design parameters with the solution is somewhat creative and original to complete the project (good values). The solution aligned with most problem statement in Chapter 1.	Deal with many components or design parameters with the solution is creative and presents 100% originality to complete the project (excellent values). The solution completely aligned with all problem	2

			statement in Chapter 1.	statement in Chapter 1.	one problem statement in Chapter 1.		statement in Chapter 1.	
A3	Investigation (15 Marks)	Not Available	Demonstrates very poor understanding of the overall research methodology, design method and data analysis techniques without explanation of the instrumentation and data collection. The limitations and delimitations related to the methodology is not provided. Design is not aligned with gap define in literature review.	Demonstrates poor understanding of the overall research methodology, design method and data analysis techniques with brief explanation of the instrumentation and data collection. The limitations and delimitations related to the methodology is not provided. Design is fairly aligned with gap define in literature review.	Demonstrates good understanding of the overall research methodology, design method and data analysis techniques with partly acceptable the explanation of the instrumentation and data collection. The limitations and delimitations related to the methodology is provided. Design is acceptably aligned with gap define in literature review.	Demonstrates very good understanding of the overall research methodology, design method and data analysis techniques with most acceptable explanation of the instrumentation and data collection with provide the limitations and delimitations related to the methodology. Design is mostly aligned with gap define in literature review.	Demonstrates in-depth understanding of the overall research methodology, design method and data analysis techniques. Describes a detailed explanation of the instrumentation and data collection with provide the limitations and delimitations related to the methodology. Design is highly aligned with gap define in literature review with additional	3

							justification provided.	
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B. CHAPTER 4: RESULT & DISCUSSION (65 MARKS) [CO2, PO4] & [CO3, PO5]

Question No.			1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
B1	Appropriately demonstrate results (Figures, tables, data etc.) (15 Marks)	Not Available	Results are poorly presented. No explanation on results or explanation of results was not relevant at all with results.	Results are appropriately presented. But explanation of results was inadequate and not neatly done. Explanation did not support comprehension of research.	Results are appropriately presented. Results and explanation included. Fair explanation and arguments.	Results of the research properly presented. Explanation on result was good, accurate and understandable.	Accurate results were well presented. Information presented supported research done and in line with arguments given. Utilize computer modeling and simulation software tools/packages/ in analyzing the data	3

B2	In-depth Analysis of results (15 Marks)	Not Available	Analysis of data was too brief, not accurate and without detailing. Arguable data collection process. Poor analysis with many major flaws in interpretation. Results presented do not answer objectives in Chapter 1.	Analysis was done with brief data collection not accurate. Detailing of analysis was unsatisfactory and disputable outcome. Weak analysis with Incorrect interpretation. Results presented weakly answer objectives in Chapter 1.	Data collection was done using normal method. Outcome of analysis was fair. Result verification acceptable under normal condition. Adequate analysis with basic interpretation. Results presented answer most objectives in Chapter 1.	Good analysis of result. Data collection method was accurately done. Detailing of analysis resulted normal verification. Competent analysis with some originality shown in interpretation. Results presented answer all objectives in Chapter 1.	Complete and in-depth analysis of research data through statistical procedures. Thorough analysis with insight and originality shown in interpretation. Results presented answer all objectives in Chapter 1 and supported with justifications.	3
B3	Concise and precise discussion (20 Marks)	Not Available	Discussion is very brief and unclear, no description on the	Discussion is somehow adequate but no description on the implication of findings.	Discussion is very in-depth and detailed, appropriately describe the	Discussion is very in-depth and detailed, moderately describe the	Discussion is very in-depth and detailed, clearly describe the	4

			implication of findings.		implication of findings.	implication of findings	implication of findings.	
B4	Modern Tool Usage (15 Marks)	Not Available	Results are acquired without usage of modern tools. Explanation of results involves no usage of modern tool.	Results are acquired with minimum usage of modern tools. Explanation of results involves minimum usage of modern tool.	Results are acquired with some usage of modern tools. Explanation of results involves some usage of modern tool.	Results are acquired with considerable usage of modern tools. Explanation of results involves considerable usage of modern tool.	Results are acquired with extensive usage of modern tools. Explanation of results involves extensive usage of modern tool.	3

C. CHAPTER 5: CONCLUSION (10 MARKS) [CO5, PO12]

Question No		0 (Not Available)	1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
C1	Conclusion (10 Marks)	Not Available	Conclusions and recommendations are available but are unclear.	Conclusions and recommendations have little relationship to the problem statements and objectives.	Conclusions and recommendations are provided with only fair research outcomes.	Conclusions and recommendations are satisfactory, complete, and relevant with the problem statements and the objectives.	Conclusions are highly significant to the problem statements and objectives of the research. The recommendations are extraordinary for advanced applications /functionality/analysis.	2

D. REPORT STRUCTURE (15 MARKS) [CO4, PO10]

Question No		0 (Not Available)	1 (Very Poor)	2 (Poor)	3 (Average)	4 (Good)	5 (Excellent)	Weightage
D1	Report organization (5 Marks)	No report available	Poorly structured with no clear progression through and between each sections. Scope of research are not considered in report writing.	Moderately structured with little progression through and between each sections. Scope of research are fairly considered in report writing.	Well-structured with acceptable progression through and between each sections. Scope of research are mostly considered in report writing.	Well-structured with good progression through and between each sections. Scope of research are always considered in report writing.	Excellent structured with clear and logical progression through and between each sections. Scope of research are always considered in report writing and being emphasized throughout the report.	1

D2	Abstract and Overall Report writing (5 Marks)	No report available	Writing is incorrect and very hard to read, numerous spelling and grammatical errors. Abstract is not relevant to the research.	Writing is somehow correct but not pleasant to read, numerous spelling and grammatical errors. Abstract is fairly relevant to research but many element not included.	Writing is basically correct and readable , spelling and grammatical errors are present but at acceptable quantities. Abstract have most required element.	Writing is correct and pleasant to read, some spelling and grammatical errors. Abstract have all required element.	Writing is fluid and pleasant to read, No spelling errors and few grammatical errors. Abstract have all required element (introduction , problem statement, objective, methodology and solution) and the writing is accurate and precise.	1
D3	Figures, tables and references format	No tables & figures included	Unacceptabl e used of figures, tables and	Poor used of figures, tables and related references	Adequate used of figures, tables and	Good used of figures, tables and related references	Excellent used of figures, tables and related	1

	(5 Marks)	in report	related references		related references	(mostly within 5 years)	references (mostly within 5 years)	
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